LAKSHAY GUPTA

SOFTWARE ENGINEERING STUDENT | BACKEND AND AI APPLICATIONS

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PROFESSIONAL SUMMARY

Second-year B.Tech Computer Science student building backend and AI-powered web applications. Hands-on experience with Python, FastAPI, SQLAlchemy, SQLite, retrieval-augmented generation, and JavaScript frontends through a medical chatbot project. Strong foundation in data structures, object-oriented programming, and C++; also experienced in designing complex gameplay systems in Unreal Engine 5.

EDUCATION

Guru Gobind Singh Indraprastha University (GGSIPU) *2024 - 2028*

B.Tech in Computer Science Engineering (Current CGPA: 8.6) *New Delhi, India*

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Computer Architecture.

DL DAV Model School, Pitampura *2024*

Class XII (CBSE) - Computer Science *New Delhi, India*

- Academics: 98% in Computer Science | 86% overall (PCM + CS).
- Achievement: Winner, Inter-School Programming Competition (Scratch Logic Building), 2021.

TECHNICAL SKILLS

Languages: Python, C++, C, C#, SQL, JavaScript, HTML/CSS.

Backend & AI: FastAPI, SQLAlchemy, LangChain, Retrieval-Augmented Generation (RAG), Google Gemini API.

Databases & Tools: SQLite, MySQL, Pinecone vector database, Git/GitHub, VS Code, Visual Studio.

Game Development: Unreal Engine 5 (Blueprints), Unity (beginner), Behavior Trees, EQS, Animation Blueprints, Blender (basic).

SOFTWARE ENGINEERING PROJECTS

RAG Mini Project: Medical Knowledge Chatbot *2026*

Python, FastAPI, LangChain, Google Gemini, Pinecone, SQLAlchemy, SQLite, JavaScript

- Built a full-stack medical-question answering application with a FastAPI backend, JavaScript frontend, and persistent user chat history.
- Implemented retrieval-augmented generation using Pinecone vector search, Google Gemini embeddings/LLM, LangChain prompts, and conversation-aware retrieval.
- Designed SQLite models for users, chats, and messages; added JWT-based login, saved answer sources, automatic chat titles, and stop-generation handling.

Smart Attendance Module (SIH Hackathon) *2024*

Python, Flask, MySQL

- Developed backend logic for a QR-based attendance tracking system.
- Integrated MySQL to automate real-time student record updates.

BanKCode: Financial Management System *2025*

C++, Object-Oriented Programming, File Handling

- Built a console-based banking application applying core object-oriented programming principles.
- Implemented file-based persistence using custom serialization for account data storage.

ADDITIONAL TECHNICAL WORK - GAME DEVELOPMENT

Advanced Gameplay Systems (Unreal Engine 5) *2024 - Present*

AI Systems, Behavior Trees, EQS, Motion Warping, Inverse Kinematics

- Designed a priority-queue-driven combat director and modular behavior trees for coordinated AI states including flanking, attacking, and staggering.
- Built tactical AI positioning with EQS, line-of-sight checks, and cover/spacing evaluation; added responsive combat input buffering and frame-accurate hit windows.
- Developed procedural traversal and locomotion systems using raycasts, sphere traces, vector math, and inverse kinematics.

Detailed gameplay video breakdowns and system logic are available in my portfolio.